STAFF REVIEW · H446 243 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. False.

2. To reduce complexity while retaining data needed to answer the question.

3. The address of the next instruction to fetch.

4. 1+1+1=3 decimal=11 binary; Sum=1, carry-out=1.

I Do — source worked steps

1. Predict the method — Write +5 in exactly 8 bits.

2. Check and explain — Invert each bit, then add one to obtain the two's complement result for −5.

We Do — guided CHECK

1. Predict the method
First calculate A AND B = 0.
Check and explain
Negate it: ¬0=1. Include intermediate reasoning when a question asks for it.

You Do — source mark schemes

1. MARK SCHEME /15. (a) +18=00010010 [1], invert 11101101 [1], add 1=11101110 [2]. (b) Sign-preserving right shift gives 11110111 [2] = −9 [1]. (c) Correct binary addition yields 10001_2 [2], correct carry evidence [1]. (d) $0.0011\times2^5 \rightarrow 0.0110\times2^4$ [2] $\rightarrow 0.1100\times2^3$ [2], decimal 6 [1].

2. MARK SCHEME /15. (a) Outputs 0,1,1,0 [4]. (b) ¬A ∨ ¬B [2], De Morgan swaps AND to OR and negates components [1]. (c) Sum=0 [1], Carry=1 [1], XOR gives sum [1], AND gives carry [1]. (d) B∧¬B=0 [2]; A∨0=A [2].

3. MARK SCHEME /15. (a) PC address of next instruction [1], MAR memory address accessed [1], MDR data/instruction transferred [1], CIR current instruction [1], ACC result/intermediate arithmetic [1]. (b) PC address into MAR [1]; instruction fetched from memory into MDR [1]; PC advanced [1]; instruction copied to CIR [1]; control unit decodes [1]; processor executes and updates registers as needed [1]. (c) Interrupt signals event requiring attention [1]; CPU checks/services it at a suitable point [1]; relevant ISR handles event [1]; execution state restored and normal processing resumes [1].

Exit Ticket — source answer

1. False. Preserve the specified width and account for overflow.